IIITB COMET FOUNDATION

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Chapter:	Determinants
Exercises:	4.3 & 4.4

EXERCISE 4.3 AND 4.4

EXERCISE – 4.3

1. Find area of the triangle with vertices at:

(a) $(1,0),(6,0),(4,3)$

(b) $(2,7),(1,1),(10,8)$

(c) $(-2,-3),(3,2),(-1,-8)$

2. Show that the points $A(a,b+c)$, $B(b,c+a)$, $C(c,a+b)$ are collinear.

3. Find values of k if area of triangle is 4 sq units and vertices are:

(a) $(k,0),(4,0),(0,2)$

(b) $(-2,0),(0,4),(0,k)$

4. (a) Find equation of line joining $(1,2)$ and $(3,6)$ using determinants.

(b) Find equation of line joining $(3,1)$ and $(9,3)$ using determinants.

5. If area of triangle is 35 sq units with vertices $(2,-6)$, $(5,4)$ and $(k,4)$, then k is:

(A) 12 (B) -2 (C) $-12,-2$ (D) $12,-2$

EXERCISE – 4.4

1. Write minors and cofactors of the elements of:

(a) $\begin{pmatrix} 2 & -4 \\ 0 & 3 \end{pmatrix}$

(b) $\begin{pmatrix} a & c \\ b & d \end{pmatrix}$

2. (a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 0 & -4 \\ 3 & 5 & 1 \\ 0 & 1 & 2 \end{pmatrix}$

3. Using cofactors of elements of second row, evaluate $\begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$

4. Using cofactors of elements of third column, evaluate $\begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix}$

5. If $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and A_{ij} are cofactors, then value of Δ is:

(A) $a_{11}A_{31} + a_{12}A_{32} + a_{13}A_{33}$

(B) $a_{11}A_{11} + a_{12}A_{21} + a_{13}A_{31}$

(C) $a_{21}A_{11} + a_{22}A_{12} + a_{23}A_{13}$

(D) $a_{11}A_{11} + a_{21}A_{21} + a_{31}A_{31}$